Aerodynamic drag force – Problem

$$F_D(in\ N) = \frac{1}{2}\rho v^2 C_D A$$

Calculate the Drag force on Audi A4 at a speed of 60 kmph

Given :

$$Frontal\ cross\ section\ area * C_d\ for\ Audi\ A4\ (internet), C_d A = 0.51\ m^2$$

$$Density\ of\ Air\ , \rho = 1.204\ kg/m^3$$

$$Velocity\ of\ car\ in \frac{m}{s}, v = 16.667 \frac{m}{s}$$

$$F_D(in\ N) = \frac{1}{2}(1.204)16.667^2 * 0.51$$

$$F_D(in\ N) = 167.5\ N$$